

Vivek Sharma

PhD Scholar

Theoretical Neuroscience | Dynamical Systems | Whole-Brain Modeling | Neuromodulation

About me

I am a PhD researcher at the Donders Institute for Brain, Cognition, and Behaviour, Radboud University, exploring how large-scale brain dynamics emerge from interacting neural systems.

My work integrates theoretical modelling, whole-brain simulations, and neurophysiological data to study neural synchrony, criticality, and how brain stimulation can influence neural dynamics.

Languages

English — Fluent
Hindi — Native

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- Personal Website

RESEARCH EXPERIENCE

2022–Present	PhD Scholar <i>Donders Institute for Brain, Cognition and Behaviour, Radboud University</i> Nijmegen, Netherlands 📍
2021–2022	Research Assistant <i>Institute for Bioengineering of Catalonia</i> Barcelona, Spain 📍
2018–2021	Project Assistant <i>National Brain Research Centre</i> Gurugram, India 📍

EDUCATION

2016–2018	M.Sc. Biochemistry <i>Jamia Millia Islamia</i> New Delhi, India 📍
2012–2015	B.Sc. (Hons) Zoology <i>University of Delhi</i> New Delhi, India 📍

SKILLS

Theoretical Neuroscience Dynamical systems Whole-brain modeling Neural dynamics
Neurophysiology EEG / MEG analysis Spectral analysis Signal processing
Neuroimaging MRI processing Image-guided stimulation planning
Neuromodulation Transcranial Ultrasound Stimulation (TUS) Stimulation protocols
Scientific computing Python MATLAB R Linux $\LaTeX$

TEACHING & MENTORING

2022–2026	Student Supervision Donders Institute for Brain, Cognition and Behaviour, Radboud University Supervision of Master's and Bachelor's thesis students and mentoring of research students during laboratory rotations.
2021–2022	Teaching Assistant Neuromatch Academy Mentored students in computational neuroscience through project-based learning.

EUROPEAN UNION RESEARCH PROJECTS

- 2022–2026** **eBRAIN-Health (Horizon Europe, Grant No. 101058516).** Development of BrainX3, an interactive platform for visualisation and exploration of multimodal brain data and whole-brain simulations supporting the eBRAIN-Health digital neuroscience infrastructure.
- 2022–2026** **AISN — Artificial Intelligence in Stroke Neurorehabilitation (Horizon Europe, Grant No. 101057655).** Identification of dynamical EEG biomarkers of stroke recovery to support prediction of motor recovery and personalised neurorehabilitation strategies.
- 2021–2022** **TVB-Cloud — The Virtual Brain Cloud (Horizon 2020, Grant No. 826421).** Contribution to the development of a cloud-based Virtual Research Environment for large-scale brain modelling and neurophysiological data analysis within the EBRAINS ecosystem.

PUBLICATIONS

- 2026** **Sharma, V.**, Tiesinga, P. H. E., & Cabral, J. (2026). *How stimulation waveform shape affects collective oscillations in the brain networks*. bioRxiv. doi:10.64898/2026.06.16.732561.
- 2026** Khalili-Ardali, M., **Sharma, V.**, Mandahar, T. S., Santos, F. P. D., Tiesinga, P., & Ramsey, N. (2026). *Electrophysiological Signature of Stroke Recovery: Investigating EEG biomarkers for prognostic insights*. bioRxiv. doi:10.64898/2026.06.01.728505.
- 2025** **Sharma, V.**, Drost, B. J. N. M., & Tiesinga, P. H. E. (2025). *Dynamical models reveal distance to criticality in ageing brain dynamics*. bioRxiv. doi:10.1101/2025.09.26.678711.
- 2025** Drost, B. J. N. M., **Sharma, V.**, & Tiesinga, P. (2025). *From computational models to clinical practice: Whole brain modelling in stroke research*. Open Research Europe, 5, 280. doi:10.12688/openreseurope.21294.1.
- 2024** Yang, M., Drost, B. J. N. M., Aviñó, D., Felici, B., Santos, F. P., Vilarrubias, R. B., **Sharma, V.**, & Verschure, P. F. M. J. (2024). *BrainX3 3.0: Advancing Neuroinformatics and Artificial Brains for Living Machines*. In *Biomimetic and Biohybrid Systems*. doi:10.1007/978-3-031-72597-5_1.
- 2023** Verschure, P. F. M. J., Santos, F. P., & **Sharma, V.** (2023). *Redefining stroke rehabilitation: Mobilizing the embodied goal-oriented brain*. Current Opinion in Neurobiology, 83, 102807. doi:10.1016/j.conb.2023.102807.
- 2023** **Sharma, V.**, Santos, F. P., & Verschure, P. F. M. J. (2023). *Patient-specific modeling for guided rehabilitation of stroke patients: the BrainX3 use-case*. Frontiers in Neurology, 14. doi:10.3389/fneur.2023.1279875.
- 2023** **Sharma, V.**, Vilarrubias, R. B., & Verschure, P. F. M. J. (2023). *BrainX3: A Neuroinformatic Tool for Interactive Exploration of Multimodal Brain Datasets*. In *Biomimetic and Biohybrid Systems*. doi:10.1007/978-3-031-39504-8_11.
- 2022** Pathak, A., **Sharma, V.**, Roy, D., & Banerjee, A. (2022). *Biophysical mechanism underlying compensatory preservation of neural synchrony over the adult lifespan*. Communications Biology, 5(1). doi:10.1038/s42003-022-03489-4.

WORKSHOPS & TUTORIALS

- 2025** Organising team member and instructor for hands-on demonstrations, Non-Invasive Brain Stimulation (NIBS) Toolkit Workshop, Donders Institute for Brain, Cognition and Behaviour, Radboud University.
- 2024** Participant, Non-Invasive Brain Stimulation (NIBS) Toolkit Workshop, Donders Institute for Brain, Cognition and Behaviour.
- 2022–2023** Tutorial instructor on visualisation and analysis of neurophysiological data, Donders Cognition, Brain, and Technology Summer School (DCBT).

CONFERENCE PRESENTATIONS

- 2026** *Ultrasound Stimulation of Lateral Geniculate Nucleus Modulates Large-Scale Neural Synchrony in Humans.* Organisation for Human Brain Mapping Annual Meeting (OHBM 2026), Bordeaux, France. Poster presentation.
- 2025** *Dynamical Models Reveal Distance to Criticality in Ageing Brain Dynamics.* Dutch Neuroscience Meeting (DNM 2025). Poster presentation.
- 2024** *Understanding Whole-Brain Network Dynamics through Targeted Stimulation and Generative Modeling.* BrainModes Conference 2024. Poster presentation.
- 2023** *Exploring the Post-Stroke Frequency-Mediated Dynamic Repertoire of Brain Activity.* Sion Stroke Recovery Conference (SSRC 2023). Poster presentation.